

Business Analysis Thinking Model Extraction

Telco Customer Churn Analysis

Mode: audit_report

Generated by Business Analysis Learning Agent Skill

Dataset: 7043 customers | Churn Rate: 26.58%

Executive Summary

This report extracts and documents the 5 reusable business analysis thinking models applied during the Telco Customer Churn analysis. Each model is presented with its data foundation, business interpretation, and a beginner-friendly learning note.

Overall Churn Rate:	26.58%
Total Customers:	7,032
Churned Customers:	1,869
Avg Monthly Charges:	\$64.80
Avg Tenure:	32.4 months
Key Finding:	Contract type is the strongest churn predictor (42.7% monthly vs 2.8% two-year)

Model A: Decomposition (Structure Breakdown)

What Was Decomposed

The overall churn rate of 26.6% was decomposed by contract type, revealing that Month-to-month customers account for 23.5% of total churn despite being only 55.1% of the customer base.

Decomposition Table

Contract Type	Share	Churn Rate	Contribution
Month-to-month	55.1%	42.7%	23.5%
One year	20.9%	11.3%	2.4%
Two year	24.0%	2.8%	0.7%
Total	100.0%	26.6%	26.6%

Business Insight

Decomposition reveals a highly concentrated problem: changing month-to-month contract terms or improving the experience of month-to-month customers would address nearly 90% of the churn problem.

Beginner Takeaway

Always decompose aggregate metrics. A 26% churn rate looks moderate, but once broken down by contract type, the real story emerges: one segment has a 43% churn rate while another has only 3%.

Model B: Segment Behavior Divergence

Segment: Tenure Groups

Comparing churn rates across tenure groups reveals extreme divergence:

0-6 months: 53.3% churn

7-12 months: 35.9%

13-24 months: 28.7%

25-48 months: 20.4%

49-72 months: 9.5%

Difference magnitude: 5.6x (newest group vs longest-tenured)

Key Insight

The 0-6 month group churns at 5.6x the rate of the 49-72 month group. Averaging these together gives 26.6%, which describes neither group well. Never trust averages alone when segments behave differently.

Business Implication

The first 6 months are critical. Resources should be concentrated here: onboarding calls, usage tips, proactive support. If a customer survives the first year, their churn probability drops by half.

Model C: Proxy Inference (Indirect Inference)

Missing Concept: Service Stickiness

"Service stickiness" - how deeply a customer is integrated into the service ecosystem - is not directly recorded in the dataset. We infer it through proxy variables.

Proxy Variable 1: Number of Add-on Services

Customers with 4+ add-on services: 16.2% churn

Customers with 0-1 add-on services: 28.8% churn

Each additional service reduces churn by approximately 3 percentage points.

Proxy Variable 2: OnlineSecurity + TechSupport Bundle

Customers with both OnlineSecurity AND TechSupport: 9.0% churn

Customers without the bundle: 29.8% churn

Why These Proxies Are Reasonable

Customers who invest in multiple services have higher switching costs. They have integrated the service into their workflow/operations. Security and support services specifically create a dependency relationship - leaving means losing configured security settings and support access.

Risk of Mis-inference

Correlation vs causation: customers who buy more services may simply be more engaged or have higher income (confounding variables). The proxy is useful for prediction but not a guarantee that forcing services on customers would reduce churn.

Model D: Constraint vs Preference

Observed Difference: Fiber Optic vs DSL Churn

Fiber optic customers churn at 41.9%

DSL customers churn at 19.0%

Two competing hypotheses explain this gap:

Hypothesis 1: Constraint (Network/Price Problem)

"Fiber optic service has quality issues or is overpriced, causing customers to leave."

Evidence: Fiber is more expensive than DSL, and price-sensitive customers may churn.

If true: Fix the network quality or adjust pricing.

Hypothesis 2: Preference (Customer Self-Selection)

"Fiber customers are inherently more likely to choose short-term contracts."

Evidence: 68.7% of fiber customers are month-to-month vs 50.6% for DSL.

Fiber optic is often chosen by customers who want flexibility or are price-comparing.

If true: Focus on contract design and bundling for fiber customers.

Assessment

Both factors likely contribute, but the data more strongly supports the preference hypothesis given the large gap in month-to-month adoption (68.7% vs 50.6%). When fiber customers do choose annual contracts, their churn drops significantly - suggesting the problem is contract structure, not network quality.

Decision Implication

Recommended action: Offer fiber customers promotional pricing on annual contracts rather than investing heavily in network infrastructure changes. Test: run a 6-month contract discount offer for fiber customers and measure churn impact.

Model E: Leverage Point Identification

Finding the Highest-Impact Segment

Leverage value = Share of affected customers x Improvement magnitude

We evaluate candidate segments:

Segment	Share	Churn	vs Base	Leverage
Month-to-month contract	55.1%	42.7%	+16.1%	0.089
First 6 months (new)	20.9%	53.3%	+26.8%	0.056
Fiber optic internet	44.0%	41.9%	+15.3%	0.067
Electronic check payment	33.6%	45.3%	+18.7%	0.063

Primary Leverage Point

Month-to-month customers (share: 55.1%, leverage: 0.089) are the highest-impact target. Improving retention of this segment by converting even 10% to annual contracts would reduce overall churn by an estimated 2-3 percentage points.

Why This Matters

Resources are finite. The leverage point heuristic ensures you invest where the return is highest. Improving the month-to-month experience benefits 55% of customers simultaneously, while fixing fiber optic alone only addresses 44% of the base.

Recommended Focus

1. Target month-to-month customers in their first 6 months (intersection of 2 high-leverage segments)
2. Offer: free OnlineSecurity + TechSupport trial for 3 months
3. Follow up: promotion to switch to annual contract at month 4
4. Measure: 6-month retention rate and add-on service adoption rate

5 Self-Check Questions

Q1: What total metric can be decomposed, and by which dimensions?

Total churn rate (26.6%) decomposed by contract type, tenure group, internet service type, payment method, family status, and add-on service count. The most informative decomposition was by contract type, revealing a 15x gap between best and worst segments.

Q2: After decomposition, how large are the subgroup behavior differences (including multiples)?

The largest gap is between tenure groups: 0-6 month customers churn at 53.3% vs 49-72 month at 9.5% - a 5.6x difference. Contract type gap: Month-to-month 42.7% vs Two year 2.8% - a 15.2x difference.

Q3: What missing key information can be inferred by a proxy variable?

"Service stickiness" inferred via add-on service count and the OnlineSecurity+TechSupport bundle. These proxies explain churn variation effectively: 0 services = 21.4% churn, 6 services = 5.3% churn.

Q4: Is the observed difference a preference or a constraint?

Fiber optic churn: primarily preference (self-selection into month-to-month contracts), not a network quality constraint. Evidence: 68.7% of fiber customers choose monthly vs 50.6% for DSL.

Q5: Which segment is both high-share and high-difference (a leverage point)?

Month-to-month customers: 55.1% of base, churn rate 42.7% (+16.1% vs average). Leverage value: ~8.9. Second priority: new customers (first 6 months).